15 factor

2026/3/26

写在前面：暂无 :)

什么是factor

- 你可以把 `factor` （因子）理解成 `R` 里**专门存放“分类 / 分组信息”**的一种特殊数据类型，不是普通数字、也不是普通文本：
  - 下图的例子， `factor` 将Group中的"A"/"B"变成了两个**标签**，分别对应这两种情况：

```
Levels: A B
> Group
  [1] "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A"
 [17] "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A"
 [33] "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A"
 [49] "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A"
 [65] "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A"
 [81] "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A" "A"
 [97] "A" "A" "A" "A" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B"
[113] "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B"
[129] "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B"
[145] "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B"
[161] "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B"
[177] "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B"
[193] "B" "B" "B" "B" "B" "B" "B" "B"
> group
  [1] A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
 [34] A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
 [67] A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
[100] A B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B
[133] B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B
[166] B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B
[199] B B
Levels: A B
>
```

- 和普通文本的区别在于：
  - 普通文本（character）：只存文字，没有顺序
  - 因子（ `factor` ）：存**分类标签**，自带有限个固定类别，还能设置顺序
  - 例如，常见的场合：
    - 性别：男、女
    - 等级：低、中、高
    - 颜色：红、绿、蓝
    - 成绩：A、B、C、D
- `factor` 有什么用呢？

- 统计分类非常方便，它的输出多了一行 `Levels` （水平 / 类别），非常直观；
- 画图自动分组，柱状图、箱线图、分组分析， `R` 只认 `factor` 当分组变量；
- 可以设置有序分类，例如，

```
# 无序 factor
level1 <- factor(c("低","中","高"))
# 有序 factor
level2 <- ordered(c("低","中","高"), levels = c("低","中","高"))
```

  这样， `R` 就知道：“低 < 中 < 高”，排序、分析都会正确。
- 节省存储空间， `factor` 内部不存重复文本，只存数字编号+标签，大数据时更高效。

factor的常用传入参数

- `ordered` ：分组的顺序；
- `levels` ：有几个分组；
- `labels` ：重命名这几个标签。
  - 例如，通过传入c("M", "W")的 `labels` 标签，可以将 `factor` 的分组重新命名。

```
R 4.5.3 · D:/Codez/R_Learning/02-系列教学1(BV1G84Uz3EgB)/01每天一点R语言（基础篇）已完结/R101/
  [1] A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
 [34] A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
 [67] A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A A
[100] A B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B
[133] B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B
[166] B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B B
[199] B B
Levels: A B
> group <- factor(Group, labels=c("M", "W"))
> group
  [1] M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M
 [34] M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M
 [67] M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M M
[100] M W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W
[133] W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W
[166] W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W W
[199] W W
Levels: M W
>
```